

HAMZA BILAL

Full-Stack MERN Developer | AI/ML Engineer

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PROFESSIONAL SUMMARY

Full-stack MERN and AI/ML developer who ships production-grade systems end to end — from MongoDB data models and Express APIs to React frontends and Python AI microservices. Sole architect of multiple complete platforms integrating large language models (Claude, Gemini, Groq) into real-world products for Pakistan's agriculture, fintech, and business-services sectors. Delivered a multi-tenant SaaS platform for a digital agency client. Seeking full-time and freelance roles building scalable, high-impact web and AI applications.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), TypeScript, Python, C/C++, Java, SQL, Bash

Frontend: React.js, Vite, React Router, Tailwind CSS, Recharts, HTML5, CSS3

Backend: Node.js, Express.js, Python FastAPI, REST API design, JWT authentication, bcrypt, role-based access control (RBAC)

Databases: MongoDB (Mongoose, Aggregation Pipeline, indexing, sharding, multi-tenancy), MySQL

AI/ML: Scikit-learn, TensorFlow/Keras, Pandas, NumPy, Decision Trees, K-Means, PCA

AI Integration: Claude & Gemini APIs (SSE streaming), Groq API, LangChain, VADER sentiment analysis

Tools: Git, GitHub, Postman, VS Code, Google Colab, Figma, npm

EXPERIENCE

Full-Stack Developer (Contract) — Zavix Digital

2026

Client Project: OpsSphere — Multi-Tenant Business Management Platform · Vehari, Punjab

- Designed and delivered **OpsSphere**, a multi-tenant MERN SaaS platform spanning **14 business modules**, enabling multiple client organizations to operate on a single, securely isolated system.
- Engineered strict tenant isolation by scoping every record and query to an organization ID, and implemented role-based access control with a granular permission catalog, standard role tiers (Owner, Admin, Manager, Client User), and support for custom roles.
- Built a hardened authentication layer using short-lived JWT access tokens with rotating refresh tokens, plus audit logging that preserves a traceable record of who changed or deleted what, and when.

Freelance Full-Stack & ML Developer — Fiverr

2026 - Present

Self-Employed · Remote

- Deliver end-to-end MERN development and ML integration for global clients, focused on production-ready architecture and clean, well-documented codebases.

PROJECTS

GreenVault — Agri-Supply Chain ERP

2026

Final Year Project (COMSATS) · MERN + Python/Flask ML Microservice · Sole Developer

- Architected a production-grade MERN ERP for Pakistan's agri/pesticide supply chain with **14 MongoDB collections and 75k+ records**; JWT (cookie + Bearer) auth, StockLog/CashBook financial tracking, auto-generated sequential IDs, and a 6-hour alert scheduler.
- Built "Crop Doctor AI" — a diagnostic module pairing MongoDB aggregation with Claude API SSE streaming to identify crop diseases and recommend pesticides in real time.
- Authored an IEEE 830-compliant SRS and an end-to-end ML pipeline (Node/Express backend + Python/Flask service) single-handedly.

PSX AI Portfolio Platform

2025

Full-Stack Investment App · React + Express + Python FastAPI + Google Gemini

- Built an AI-driven platform for Pakistani retail investors: portfolio tracking, risk analysis, stock comparison, and a Gemini-powered chat assistant covering **68 PSX stocks across 18 sectors**.
- Engineered an 8-factor recommendation engine (P/E, dividend yield, ROE, beta, volatility, risk & goal alignment) plus a FastAPI service computing Sharpe ratio, volatility, and VaR from live yfinance prices.
- Implemented secure auth (JWT httpOnly cookies, email-OTP verification, password reset), RSS-based market news with VADER sentiment, and a themeable dark/light React UI.

ContentForge — AI Content Generator

2024

Personal / Freelance SaaS · React + Express + Groq API

- Built an AI SaaS generating SEO blog posts, social captions, and product copy on demand, with a Tailwind UI, JWT-protected routes, and MongoDB-stored generation history.

EDUCATION

Bachelor of Science in Computer Science (BSCS)

2023 - 2027 (expected)

COMSATS University Islamabad, Vehari Campus · CGPA: 3.71 / 4.0

Relevant coursework: Machine Learning, Natural Language Processing (NLP), Advanced Database Systems, Advanced Web Technologies, E-Commerce & Digital Marketing, Theory of Automata, Data Structures, Object-Oriented Programming.